

Failure log

No real room was available. The run below is the live-agent demo on the synthetic library (submission/live_demo_run, the run in the demo video): 61 books for 60, titles 98.2% right with one confidently wrong, 17 of 20 spines within 15%, 118 s to packet.

The three worst errors in the demo run

1. One box over two spines gave its title to the wrong book (the one confidently wrong title)

- **What happened.** *Five Point Someone* appears twice (B006, B015, both at confidence 1.0). B006 is really the unreadable spine to its left: in frame f0001 the model drew one box over both spines (27.5–31.6 cm) and labelled it *Five Point Someone*; that box joined B006's column, and a labelled reading beats the unlabelled ones, so B006 took the title. It also shows up as a 30% "height error": B006's 20.2 cm is compared with the real book's 29 cm.
- **Root cause.** A box twice as wide as the spines it overlaps is a merge of two spines, but it still counted as a full sighting of one of them, reading included.
- **What I would change.** Treat a box that spans two established columns as untrusted for both its geometry and its reading (as boxes placed by text already are), and let a title only name a column where it fits the column's width.

2. A box around a label inside a spine shortened *Clean Code* by 35%

- **What happened.** In frame f0000 the model returned two boxes for *Clean Code*: the whole spine (1.6–32.1 cm) and a small one inside it (23.7–31.0 cm, a label). Both counted as uncut sightings, so the spine's top became the median of 1.6 and 23.7: 19.5 cm reported for 30.2 cm.
- **Root cause.** Two boxes from the same frame were allowed to be two sightings of one book.
- **What I would change.** One sighting per book per frame: a box lying inside another box from the same frame is a part of that spine and is dropped.

3. An author's name read as a title, identified title-only (*Susan Cain*)

- **What happened.** On a spine whose title was outside the frame, the model returned "Susan Cain" as the title with no author; a catalog record titled that way matched title-only.
- **Root cause.** The title-equals-author guard only fires when both fields are present.
- **What contained it.** Title-only matches are capped at 0.7, so B026 went to the review queue and out of the claim total instead of being confidently wrong.
- **What I would change.** Check a lone "title" against the catalog's author names before accepting it as a title.

Cost per sweep and latency per stage

Recorded in each packet under `stages` (`latency_s`, `usage`, `cost_usd_estimate`). The live agent demo (submission/live_demo_run): 16 spine and 6 item vision calls (about 29k input and 24k output tokens, estimated US\$0.07 at paid Flash-Lite rates; free tier used), 1 live price search and 99 from cache, and after the sweep: 65 s finishing the vision backlog, 49 s identification, under 3 s pricing.

The development failures found and fixed on the way (17, each with root cause and measured effect) are in `docs/FAILURE_LOG.md` in the repository.